Aakash Kumar

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EDUCATION

Northeastern University - MS in AI, GPA: 3.7/4 **Boston, MA | 2026 - Present**
Relevant Coursework: AI, Data Analytics, ML Statistics, MLOps, Data Mining

VIT Chennai - B.Tech in CSE with specialization in AI & ML, CGPA: 9.35/10 **Chennai, India | 2021 - 2025**
Relevant Coursework: DSA, Deep Learning, NLP, Computer Vision, Reinforcement Learning, LLMs

SKILLS

Programming	Python, C++, SQL, PostgreSQL, MS Office, HTML, CSS
LLM/GenAI	RAG Pipelines, GraphRAG, Agentic AI, LangChain, LangGraph, Pydantic AI, Multi-agent Architectures, Vector Databases (Pinecone, ChromaDB), LLM Fine-tuning (QLoRA/LoRA), Prompt Engineering, Transformers, Hugging Face, Gemini API, LLM Observability (LangSmith)
Machine Learning	PyTorch, TensorFlow, Scikit-learn, Deep Learning, Federated Learning, Model Training, Model Evaluation
MLOps/Deployment	Docker, CI/CD, MCP, REST APIs, FastAPI, n8n, Model Deployment, Workflow Automation, Azure (ML Studio, Blob Storage, Databricks), GCP, Git/GitHub, Terminal, Agile/Scrum
Data Engineering	Neo4j, ETL Pipeline Development, Pandas, NumPy, Web Scraping
Developer Tools	Cursor, GitHub Copilot, Claude Code
Interpersonal	Leadership, Team Management, Project Management, Time Management, Communication, Problem Solving, Cross-Functional Collaboration, Adaptability

Experience

AI Automation Intern **Divine Success Flow**

- Built **15+ end-to-end AI automation workflows** using **n8n**, integrating Telegram, WhatsApp CRM, Pinecone, and multi-step data pipelines processing **500–2,000 records per run** with average execution latency under **8 seconds**.
- Developed and deployed **LLM-based agents** and **RAG pipelines** using OpenAI Agent Builder and Gemini for lead generation and CRM automation, achieving **90%+ lead qualification accuracy** over a **200K-vector Pinecone index** with **sub-50ms retrieval latency**.
- Automated scraping (**1,000+ records/hour**), transcription (**Whisper, ∼15 sec/audio-min**), text-to-speech, and content workflows, reducing manual effort by **∼70%** across **5 client-facing operational pipelines**.

AI Intern **Modern Agriculture Technology Innovation Center**

- Developed a mobile app for farm-yield prediction and farmer decision support, delivering end-to-end inference results in under **400 ms** for real-time field usability.
- Collected and manually annotated **3,000+ farm-field images** with bounding boxes across **4 ripeness classes**, expanding to **8,000+ samples** via augmentation (flip, rotate, brightness, crop).
- Applied **YOLOv11n** (**mAP@0.5: 0.91**, **∼6 MB** model) for mango detection and **DenseNet121** for ripeness classification, achieving **∼92% accuracy** and **0.92 F1-score** with **sub-50ms on-device inference**.

Academic Support Intern & Computer Science Instructor Akash EM High School
Operations & Teaching Role *2024 – 2025*

- Managed student attendance systems and automated WhatsApp notifications for absences and fee updates, improving school–parent communication.
- Prepared and formatted academic materials, including question papers and syllabus documentation for faculty use.
- Taught Python and HTML to primary students through interactive sessions, introducing foundational programming concepts.

TEACHING ASSISTANTSHIP

- **Teaching Assistant** - *Responsible AI (PHIL 5110)* **Northeastern University | Fall A 2026**
  - Graded and delivered detailed, rubric-based feedback for 90+ students across recurring "Responsible AI in the Wild" essays and reflective exit tickets, calibrating scoring against instructor-provided benchmarks for consistency.
  - Identified and flagged conceptual gaps, academic integrity concerns, and conflicts of interest in student submissions, escalating nuanced cases to faculty for review.
  - Responded to student grade inquiries and feedback disputes with clear, evidence-based explanations, revising assessments where warranted to ensure fairness.
  - Maintained organized, roster-aligned gradebooks with per-criterion scoring to support transparent and defensible grading across multiple submission batches.
- **Teaching Assistant** - *AI in Healthcare (HLTH 5800)* **Northeastern University | Summer B 2026**
  - Graded and delivered detailed, rubric-based feedback for 45+ students across 7 modules spanning AI applications, algorithmic bias, regulatory compliance (HIPAA/GDPR), and CHAI-standard model card documentation.
  - Evaluated discussion contributions against a multi-criterion rubric, providing individualized feedback to support continuous student improvement.
  - Managed student communications and resolved grading queries via HelpScout and email, while flagging academic integrity concerns and conceptual gaps to faculty.
  - Maintained organized gradebooks to ensure consistent and transparent grading standards throughout the semester.
- **Teaching Assistant** - *Workplace Skills in Action (EEBA 6410)* **Northeastern University | Summer A 2026**
  - Supporting graduate students in navigating course concepts, resolving doubts, and building practical workplace competencies.
  - Facilitating an open and approachable environment to encourage active student participation and early clarification of queries.
  - Leveraging course knowledge to provide timely, contextual guidance while gaining new perspectives through student-driven discussion.
- **Teaching Assistant** - *Negotiations (MGMT 6214)* **Northeastern University | Spring 2026**
  - Stepped into the TA role mid-semester for the final three weeks, rapidly orienting to course material and student needs.
  - Supported students with course content and assignment guidance, developing an ability to quickly assess and address knowledge gaps.
  - Strengthened adaptability and responsiveness in a fast-paced academic support setting.

Projects

TrialMatch: Privacy-Preserving Multi-Agent Clinical Trial Matching
LangGraph, Ollama, Hybrid RAG, BM25, Gemini API, pytest, Streamlit, TREC 2021

- Architected a privacy-preserving **multi-agent** clinical trial screening system in **LangGraph**, orchestrating a six-stage pipeline (patient profiler, pre-filter, rule checker, eligibility checker, critic, ranker) with stateful graphs and conditional routing, running a local **Qwen2.5-3B** model via **Ollama** on a CPU-only 8 GB laptop so patient notes never leave the machine.
- Enforced a two-tier **privacy boundary in code**: only public ClinicalTrials.gov text reaches the cloud **Gemini API** through a single type-guarded module that rejects any patient data type, verified by an automated **pytest** suite that also scans the codebase to confirm no other module imports a cloud SDK.
- Built an offline data pipeline ingesting **TREC 2021 Clinical Trials** patient topics and expert relevance judgments plus **ClinicalTrials.gov API v2** records, parsing **1,927 eligibility criteria** from 118 trials into atomic inclusion/exclusion rules with **71 hand-verified numeric thresholds** (age, eGFR, HbA1c, LVEF, ECOG) evaluated deterministically without the LLM.

- Developed a **hybrid RAG** layer fusing **BM25** and **nomic-embed-text** dense retrieval with **Reciprocal Rank Fusion** over 237 indexed trials, plus a **44-card medical knowledge base** (drug classes, disease groups, staging systems) retrieved per criterion, raising ranking quality to **0.831 nDCG@10** and **0.475 P@5** vs **0.712** and **0.300** for BM25 on held-out patients.
- Designed an **evidence-grounding guard** and **critic agent** that reject decisions not supported by a verified span of the patient note, blocking invented values and fabricated denials; diagnosed and fixed a profiler failure where the 3B model missed a primary diagnosis and hallucinated a negative finding.
- Engineered for constrained hardware with **JSON-schema constrained decoding**, validation-driven retries, prompt-prefix ordering for KV-cache reuse, a 2,048-token output cap to stop runaway generation, and a deterministic disk cache that makes multi-day evaluation runs exactly resumable after crashes.
- Ran a **pre-registered evaluation** with a dev/test split, **SHA-256 code freeze**, paired bootstrap 95% CIs (10,000 resamples) and exact Wilcoxon signed-rank tests across five systems, including single-call, no-critic and RAG ablations on 96 held-out patient-trial pairs.
- Diagnosed through error analysis of **1,252 criterion decisions** that grounding constraints left **92%** of criteria undecided (506 of 553 LLM answers lacked verifiable evidence), explaining why a single-call baseline ranked best (**0.882 nDCG@10**) and identifying evidence quoting, cohort logic and temporal reasoning as the key bottlenecks.
- Shipped an offline **Streamlit** demo that screens currently recruiting Massachusetts trials, streams each agent's progress live, and renders a criterion-by-criterion eligibility checklist with quoted evidence for clinician review.

Concrete Crack Detection Using a Hybrid Deep Learning Framework (Patent: 410906-001)

- Proposed a hybrid deep learning framework for automated concrete crack detection and severity assessment, achieving **96.8% classification accuracy** and **0.98 AUC-ROC** across binary and multi-class crack severity tasks.
- Formulated a multi-model feature extraction pipeline using pretrained CNNs (InceptionV3, ResNet50, DenseNet, VGG), fusing ∼**9,216-dimensional** feature vectors with an average extraction time of **18–29 ms per image on GPU**.
- Transformed deep features into vector representations and applied machine learning models (SVM, XGBoost) for crack classification, achieving **97% F1-score** with classifier inference under **2 ms**.
- Optimized end-to-end inference pipeline to estimate crack depth within ±**0.4 mm margin of error** in under **five seconds**, enabling rapid structural health assessment across **3,000+ images**.

Remote Human-in-the-Loop Approval System for AI Coding Agents
Python, FastAPI, REST, ntfy, Pydantic, Docker

- Designed a **human-in-the-loop** approval layer for **Claude Code**, intercepting agent permission requests (shell commands, file edits) through a **PermissionRequest hook** and routing them to a mobile device for real-time Approve/Deny decisions.
- Built a **FastAPI REST relay** with resource-oriented endpoints, API-key authentication, request TTL expiry, idempotent decisions (409 on duplicates), and rate limiting, using **asyncio**-based **long-polling** to deliver decisions without inbound network access.
- Integrated **pub/sub push notifications** (ntfy) with interactive HTTP action buttons, sending decisions from the phone back to the agent in seconds across a restricted outbound-only network.
- Engineered a **fail-safe** hook in standard-library Python that never auto-approves: timeouts, network errors, and relay outages fall back to the local terminal prompt, with secret redaction applied before any data leaves the machine.
- Wrote **pytest** suites covering authentication, long-poll timeouts, expiry, and failure paths with mocked services, and containerized the relay with **Docker** for cloud deployment.

Luminary: Cross-University Research Discovery Platform (Protothon 8.0 Winner)

- Architected a privacy-preserving federated research discovery platform using **Federated Learning** to transfer model weights across institutions without exposing raw training data, enabling cross-university collaboration at scale.
- Implemented a **Quantum Encoding** pipeline to locally encode text, images, and dataset metadata into quantum states at each institution, ensuring zero raw data transmission while maintaining semantic searchability.
- Built a hybrid semantic search engine using **SciBERT embeddings** and **Cosine Similarity** for scientific-domain-aware retrieval, integrated with **Quantum Approximate Optimization Algorithm (QAOA)** to compute optimal researcher collaboration scores across federated nodes.

- Designed a **Multi-Agent AI** system comprising a **Data Bot** and a **Paper Bot**, triggered in parallel from a single user prompt: Data Bot surfaces relevant datasets while Paper Bot retrieves related research papers, delivering unified real-time results through coordinated agent execution.
- Delivered an end-to-end research discovery system reducing collaboration discovery time from **8 weeks to minutes**, eliminating research duplication across PhD-level institutional knowledge silos spanning **187+ R1 universities**.

RRF-Infused Retrieval-Augmented Generation for Text Summarization

- Proposed an RRF-infused RAG framework to improve retrieval accuracy and contextual relevance in LLM-based summarization, improving Mean Reciprocal Rank (MRR) by **19%** over standard RAG baselines.
- Engineered a hybrid retrieval pipeline using Sentence Transformers (`all-MiniLM-L6-v2`, **384-d embeddings**) and Reciprocal Rank Fusion for multi-query aggregation, achieving **Precision of 0.84** and **Recall of 0.81**.
- Implemented document ingestion, vector storage, and cosine similarity–based retrieval using PyPDF2 and SQLite, supporting a corpus of **500+ documents (∼2M tokens)** with **512-token chunks** and **50-token overlap**.
- Evaluated summarization quality using ROUGE-1, ROUGE-2, and ROUGE-L, achieving scores of **0.52 / 0.27 / 0.49** respectively, outperforming standard RAG baselines by **+9–11 points** across all metrics.

Epilepsy Classification - FCD Lesion Segmentation & Report Generation

- Curated and preprocessed **170 MRI scans** (85 healthy, 85 FCD-II patients) using the **FSL** library (skull-stripping, RAS reorientation), extracting **320 slices per volume** to generate **27,200 FLAIR images** with matched ROI masks for model training.
- Designed an **Attention-Enhanced ConvLSTM** segmentation architecture for FCD lesion detection, achieving **IoU: 0.77**, **Dice: 0.87**, and **Sensitivity: 0.8154**, substantially outperforming state-of-the-art baselines (GLFF Dice: 0.408, TBFEF-Net Dice: 0.597).
- Built a subject-identification classifier with **EfficientNetB0**, achieving **96% accuracy**, **97% macro precision**, and **96% F1-score** across **85 subject classes**.
- Integrated segmentation and classification outputs into a **RAG pipeline** powered by **Gemini LLM** to automatically draft clinical reports from lesion and patient metadata, scoring **BLEU: 0.77**, **ROUGE-1: 0.89**, and **METEOR: 0.90** against expert-authored summaries.

Detecting Stress Levels Based on Sleeping Posture

- Developed a system to detect stress levels based on sleeping posture during overnight sleep cycles.
- Implemented MediaPipe framework to analyze body posture and compute joint-angle measurements for posture classification.
- Automated stress-level estimation and generated personalized morning feedback for healthcare stress monitoring.

WhatsApp-Based School Attendance and Fee Notification System

- Developed a Python Tkinter desktop app to automate school attendance alerts and fee reminder notifications via WhatsApp Web.
- Integrated OTP-based email authentication using SMTP and SSL; processed Excel-based student records using Pandas to dynamically identify absentees and fee due dates.
- Automated message delivery using PyAutoGUI and Pyperclip, enabling real-time parent communication.

AI-Powered Stock Analysis Telegram Bot

- Designed and deployed an automated Telegram bot using n8n to deliver real-time stock chart analysis on user request.
- Integrated Twelve Data API for symbol lookup and Chart-Img (TradingView) for high-resolution stock chart image generation.
- Implemented an LLM-based analysis pipeline (Google Gemini) to interpret stock charts and generate concise insights on trends, patterns, and key levels.

Multi-Modal Brain Image Fusion Using Multi-Feature Guided Fusion Network (Publication)

- Co-authored **MFG-FuseNet**, a CNN-based fusion architecture for MRI–PET brain images, combining a **Feature Extraction Framework** (multi-level differential feature extraction across 3 convolutional blocks) with a **Feature Reconstruction Framework** (deconvolutional up-sampling guided by those feature differences) to preserve both structural and functional detail.

- Designed a **Spatial & Edge Extraction Framework (SEEF)** using a Laplacian-operator convolution block to isolate edge/texture information, and a **Context Boosting Framework (CBF)** with a 1000-filter convolutional layer to enhance deep textural features at each fusion stage.
- Trained on **260 registered MRI–PET image pairs** (256×256) from the Harvard Medical brain imaging dataset using **PyTorch/Keras** on Colab (Tesla T4/P100 GPUs), optimized with a custom **SSIM + L2-norm** loss function balancing spatial and texture fidelity.
- Achieved $\mathbf{Q^{AB/F}}$**: 0.54836**, **UQI: 0.85435**, and **RMSE: 0.10582**, outperforming state-of-the-art baselines (FDGNet, FunFuseAn, PCA, DWT) with a **30.51% improvement in** $\mathbf{Q^{AB/F}}$, **5.81% in UQI**, and a **3.81% reduction in RMSE**; published in *Biomedical Signal Processing and Control* (Elsevier).

Publications

- Sugarcane Leaf Classification using VGG19 - *Sadhana, Springer.* link
- Flowing Asanas: Enhancing Yoga Posture Detection with BiLSTM Attention and MediaPipe - *IEEE.* link
- Enhancing Diabetes Mellitus Prediction: Hybrid Deep Learning Model - *Adroidjournals.* link
- Classification and Validation of Tomato Leaf Disease Using Deep Learning Techniques - *Adroidjournals.* link
- Multi-modal Brain Image Fusion Using Multi-feature Guided Fusion Network - *Biomedical Signal Processing and Control.* link
- Classification of Healthy and Unhealthy Leaf Disease Using Modified MobileNetV2 - *IEEE Int. Conf. on Data Intelligence & Cognitive Informatics.* link
- Yawning Detection for Cognitive Distraction in Drivers Using AlexNet - *Elsevier ICECMSN.* link
- Deep Learning for Wearable Devices (Book Chapter 7) - *Elsevier.* link
- Comprehensive Analysis of Seizure Activity in Critical Care Patients with Harmful Brain Patterns (Book Chapter 11) - *CRC Press, Taylor & Francis.* link
- Deep learning based Mango Yield Prediction and Classification Enhancement using YOLOv11L and DenseNet121 Architecture - *IEEE.* link

Conference Presentations

- Air Quality Prediction - *IICAQM 2023, IISc Bangalore*
- Multi-Class Ovarian Cancer Classification: An Ensemble Approach Using Histopathological Data - *IEEE – ISBI*
- Deep Learning based Mango Yield Prediction and Classification Enhancement using YOLOv11L Architecture - link

Certifications

Advanced C++ Programming - *IIT Bombay, 2022*

Python Programming (Version 3.4.3) - *IIT Bombay, 2022*

E-Business - *NPTEL, 2023*

Achievements

- Awarded Top-Performing Science Student distinction at the school level.
- Ranked among the Top 4 students in the 12th Grade Board Examinations.
- Served as Head Prefect during secondary school, demonstrating leadership and discipline.
- Elected President, Robotics Club - VIT Chennai, leading technical initiatives and student teams.
- Achieved Top 1% ranking in the NPTEL Certification Course on E-Business.
- Acted as a Resource Person for an Online International Short-Term Training Program conducted by Scholar's Gate.

Co-Curricular Activities

President, Robotics Club VIT Chennai

- Led a student organization of 400+ members structured across 7+ departments including Teaching, Technical, Outreach, Marketing, and Operations.
- Planned, organized, and executed multiple seminars, workshops, and training programs, mentoring 120+ students per session.
- Coordinated cross-functional teams to ensure effective knowledge transfer, event execution, and sustained member engagement.

Head Prefect (Discipline Committee) Secondary School

- Headed the Discipline Committee, ensuring order and adherence to institutional policies across all academic and extracurricular activities.
- Supervised and coordinated prefects from every class to maintain discipline during daily school operations and major events.
- Acted as the primary liaison between school administration and the student body for discipline-related matters.

Resource Person Scholar's Gate

International Training Program

- Delivered multiple technical sessions and live demonstrations to PhD scholars and working professionals.
- Explained and demonstrated practical applications of Deep Learning for automating and optimizing everyday professional workflows.
- Received positive feedback for simplifying complex AI concepts into actionable, real-world solutions.